



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Simultaneous Equations

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Your notes

Linear Simultaneous Equations

Linear Simultaneous Equations

What are linear simultaneous equations?

- When there are **two unknowns** (x and y), we need **two equations** to find them both
 - For example, $3x + 2y = 11$ and $2x - y = 5$
 - The values that work are $x = 3$ and $y = 1$
- These are called **linear simultaneous equations**
 - Linear** because there are no terms like x^2 or y^2

How do I solve linear simultaneous equations by elimination?

- Elimination** removes one of the variables, x or y
- To eliminate the x 's from $3x + 2y = 11$ and $2x - y = 5$, make the number in front of the x (the **coefficient**) in both equations the same (the sign may be different)
 - Multiply **every term** in the **first** equation **by 2**
 - $6x + 4y = 22$
 - Multiply **every term** in the **second** equation **by 3**
 - $6x - 3y = 15$
 - Subtracting** the second equation from the first **eliminates x**
 - When the **sign** in front of the term you want to eliminate is the **same**, **subtract** the equations

$$\begin{array}{r} 6x + 4y = 22 \\ - \quad 6x - 3y = 15 \\ \hline 7y = 7 \end{array}$$

- The y terms have become $4y - (-3y) = 7y$ (be careful with **negatives**)
 - Solve** the resulting equation to find y
 - $y = 1$
- Then **substitute** $y = 1$ into one of the **original** equations to find x



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- $3x + 2 = 11$, so $3x = 9$, giving $x = 3$
- Write out **both solutions** together, $x = 3$ and $y = 1$
- **Alternatively**, you could have eliminated the y 's from $3x + 2y = 11$ and $2x - y = 5$ by making the **coefficient** of y in both equations the same
 - Multiply every term in the second equation by 2
 - **Adding** this to the first equation **eliminates y** (and so on)
 - When the **sign** in front of the term you want to eliminate is **different**, **add** the equations

$$\begin{array}{r} 3x + 2y = 11 \\ + \quad 4x - 2y = 10 \\ \hline 7x \quad = 21 \end{array}$$

How do I solve linear simultaneous equations by substitution?

- **Substitution** means substituting one equation into the other
 - This is an **alternative** method to **elimination**
 - You can still use elimination if you prefer
- To solve $3x + 2y = 11$ and $2x - y = 5$ by substitution
 - **Rearrange** one of the equations into $y = \dots$ (or $x = \dots$)
 - For example, the second equation becomes $y = 2x - 5$
 - **Substitute** this into the first equation
 - This means **replace** all y 's with $2x - 5$ in brackets
 - $3x + 2(2x - 5) = 11$
 - **Solve** this equation to find x
 - $x = 3$
 - Then **substitute** $x = 3$ into $y = 2x - 5$ to find y
 - $y = 1$

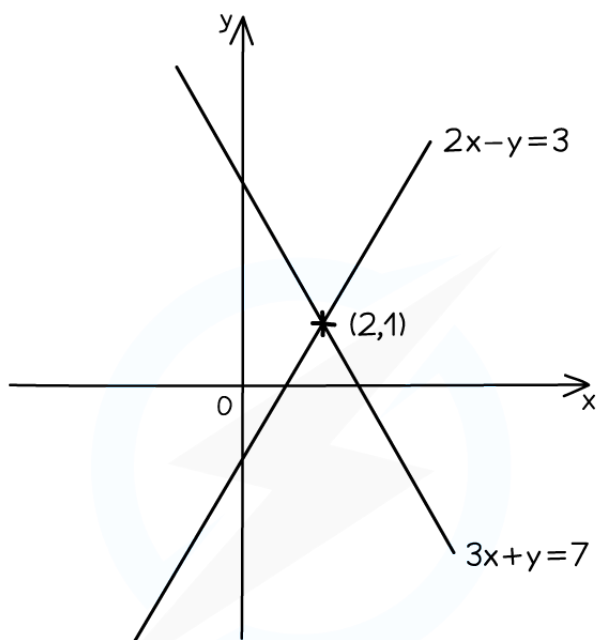
How do I solve linear simultaneous equations graphically?

- **Plot** both equations on the same set of axes
 - To do this, you can use a **table of values**

- or rearrange into $y = mx + c$ if that helps
- Find where the lines **intersect** (cross over)
 - The x and y **solutions** to the simultaneous equations are the x and y **coordinates** of the point of **intersection**
- For example, to solve $2x - y = 3$ and $3x + y = 4$ simultaneously
 - First **plot** them both on the **same axes** (see graph)
 - Find the **point of intersection**, (2, 1)
 - The solution is $x = 2$ and $y = 1$



Your notes



- LINES INTERSECT AT (2,1)
- SOLVING $2x - y = 3$ AND $3x + y = 7$ SIMULTANEOUSLY IS $x = 2$, $y = 1$

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Your notes

Examiner Tips and Tricks

- Always check that your final solutions satisfy **both** original simultaneous equations!
- Write out both solutions (x and y) together at the end to avoid examiners missing a solution in your working



Worked Example

Solve the simultaneous equations

$$5x + 2y = 11$$

$$4x - 3y = 18$$

It helps to number the equations

$$5x + 2y = 11 \quad (1)$$

$$4x - 3y = 18 \quad (2)$$

We will choose to eliminate the y terms

Make the y terms equal by multiplying all parts of equation 1 by 3 and all parts of equation 2 by 2

$$15x + 6y = 33 \quad (3)$$

$$8x - 6y = 36 \quad (4)$$

The 6y terms have different signs, so they can be eliminated by adding equation 4 to equation 3

$$\begin{array}{r} 15x + 6y = 33 \\ + \quad 8x - 6y = 36 \\ \hline 23x \quad = 69 \end{array}$$

Solve the equation to find x (divide both sides by 23)

$$x = \frac{69}{23} = 3$$

Substitute $x = 3$ into either of the two original equations

$$(1) \quad 5(3) + 2y = 11$$



Your notes

Solve this equation to find y

$$\begin{aligned}15 + 2y &= 11 \\2y &= 11 - 15 \\2y &= -4 \\y &= \frac{-4}{2} = -2\end{aligned}$$

Substitute $x = 3$ and $y = -2$ into the other equation to check that they are correct

$$\begin{aligned}\textcircled{2} \quad 4x - 3y &= 18 \\4(3) - 3(-2) &= 18 \\12 - (-6) &= 18 \\18 &= 18\end{aligned}$$

Write out both solutions together

$$\mathbf{x = 3, \quad y = -2}$$

This question can also be done by eliminating x first (multiplying equation 1 by 4 and equation 2 by 5 then subtracting)

How do I form simultaneous equations?

- Introduce **two letters**, x and y , to represent **two unknowns**
 - Make sure you know exactly what they stand for (and any units)
- Create **two different equations** from the words or contexts
 - 3 apples and 2 bananas cost \$1.80, while 5 apples and 1 banana cost \$2.30
 - $3x + 2y = 180$ and $5x + y = 230$
 x is the **price** of an apple, in **cents**
 y is the **price** of a banana, in **cents**
 - This question could also be done in dollars, \$
- Solve** the equations **simultaneously**
- Give answers **in context** (relate them to the story, with units)
 - $x = 40, y = 30$
 - In context: an apple costs 40 cents and a banana costs 30 cents



Your notes

- Some questions don't ask you to solve simultaneously, but you still need to
 - Two numbers have a **sum** of 19 and a **difference** of 5, what is their **product**?
 - $x + y = 19$ and $x - y = 5$
 - Solve simultaneously to get $x = 12, y = 7$
 - The product is $xy = 12 \times 7 = 84$



Examiner Tips and Tricks

- Always check that you've answered the question! Sometimes finding x and y isn't the end
 - E.g. you may have to state a conclusion



Worked Example

At a bakery a customer pays £9 in total for six bagels and twelve sausage rolls.

Another customer buys nine bagels and ten sausage rolls, which costs £12.30 in total.

Find the cost of 5 bagels and 15 sausage rolls.

The two variables are the price of bagels, b , and the price of sausage rolls, s

Write an equation for the first customer's purchases, and label it equation 1

$$\textcircled{1} \quad 6b + 12s = 9$$

Write an equation for the second customer's purchases, and label it equation 2

$$\textcircled{2} \quad 9b + 10s = 12.3$$

We will choose to eliminate the b terms

Make the b terms equal by multiplying all parts of equation 1 by 3 and all parts of equation 2 by 2

Label these as equations 3 and 4

$$\textcircled{1} \times 3 \quad 18b + 36s = 27 \quad \textcircled{3}$$

$$\textcircled{2} \times 2 \quad 18b + 20s = 24.6 \quad \textcircled{4}$$



Your notes

To eliminate b , subtract equation 4 from equation 3

$$\textcircled{3} - \textcircled{4} \quad 16s = 2.4$$

Solve for s

$$s = \frac{2.4}{16} = 0.15$$

Substitute this into either equation to find b , we will use equation 1

$$\begin{aligned}\textcircled{1} \quad 6b + 12(0.15) &= 9 \\ 6b + 1.8 &= 9 \\ 6b &= 7.2 \\ b &= 1.2\end{aligned}$$

So sausage rolls cost £0.15 each and bagels cost £1.20 each

Use these values to find the price of 5 bagels and 15 sausage rolls

$$(5 \times 1.2) + (15 \times 0.15) = 8.25$$

£8.25



Your notes

Quadratic Simultaneous Equations

Quadratic Simultaneous Equations

What are quadratic simultaneous equations?

- When there are two unknowns (e.g. x and y) in a problem, we need two equations to be able to find them both; these are called **simultaneous equations**
- If there is an x^2 or y^2 or xy in one of the equations then they are **quadratic** (or **non-linear**) simultaneous equations

How do I solve quadratic simultaneous equations?

- Use **substitution**
 - **Substitute** the **linear** equation, $y = \dots$ (or $x = \dots$), into the quadratic equation
 - Do not try to substitute the quadratic equation into the linear equation
- E.g. To solve $x^2 + y^2 = 25$ and $y - 2x = 5$
 - **Rearrange** the **linear** equation into $y = 2x + 5$
 - **Substitute** this into the quadratic equation, replacing all y 's with $(2x + 5)$
 - $x^2 + (2x + 5)^2 = 25$
- Expand and **solve** this **quadratic** equation
 - $x^2 + 4x^2 + 20x + 25 = 25$
 - $5x^2 + 20x = 0$
 - $5x(x + 4) = 0$
 - $x = 0$ and $x = -4$
- Substitute **each** value of x into the **linear** equation, $y = 2x + 5$, to find the corresponding y values
 - $y = 2(0) + 5 = 5$
 - $y = 2(-4) + 5 = -3$
- Present your solutions in a way that makes it obvious which x belongs to which y

- $x = 0, y = 5$ or $x = -4, y = -3$
- **Check** your final solutions satisfy both equations



Your notes

What if the quadratic has repeated roots or no roots?

- If the resulting quadratic after substituting has a **repeated root**,
 - then the line is a **tangent** to the curve
 - i.e. the curve and the line intersect in one place only
 - There is only **one solution** for x and y
- If the resulting quadratic to be solved has **no roots**,
 - then the line does not intersect with the curve
 - There are **no solutions** to the simultaneous equations
 - If this happens it *may* be an indicator that your working is wrong!

What if I can't substitute one equation into the other straight away?

- If the linear equation is **not** in the form $y = \dots$ or $x = \dots$
 - You will need to **rearrange** it first, so that it can be **substituted** into the quadratic equation
- Consider solving $xy = 3$ and $x + y = 4$
- Either:
 - Rearrange the second equation to $y = 4 - x$ and substitute into $xy = 3$
 - $x(4 - x) = 3$
 - Expanding produces a quadratic that can be solved for x
 - $4x - x^2 = 3$
 - Or rearrange the first equation to $y = \frac{3}{x}$ and substitute into $x + y = 4$
 - $x + \frac{3}{x} = 4$
 - Multiplying both sides by x produces a quadratic that can be solved for x

- $x^2 + 3 = 4x$

How do I use a graph to solve quadratic simultaneous equations?

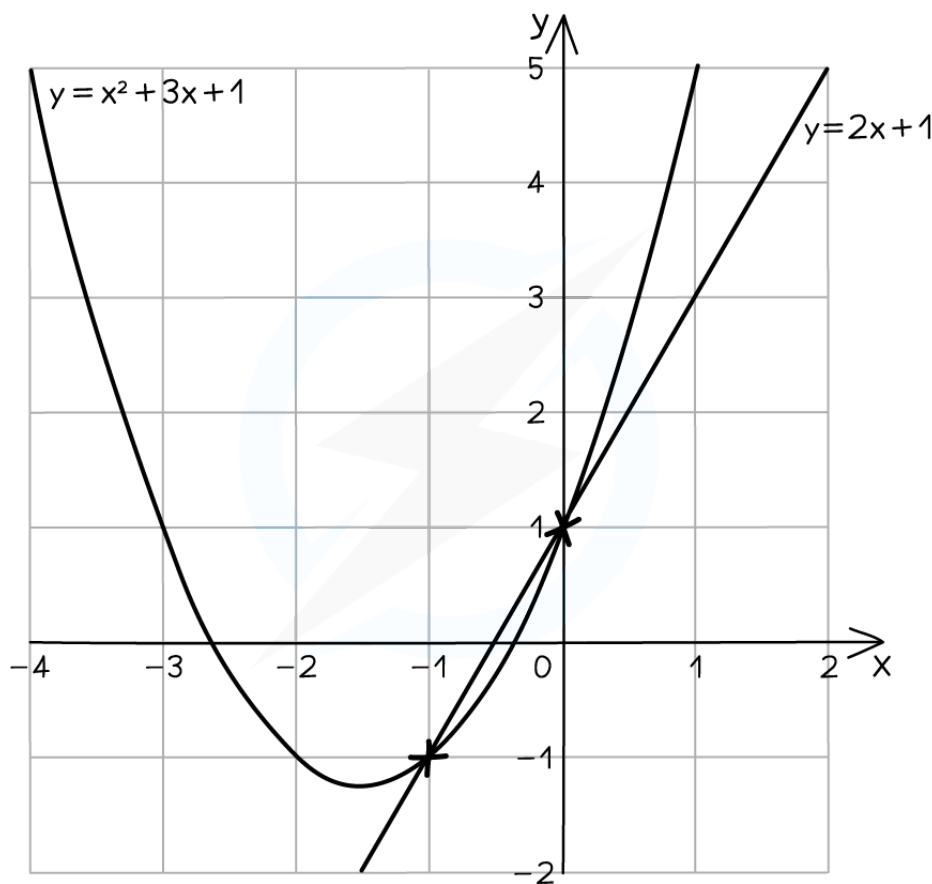
- **Plot** both equations on the same set of axes
 - To do this, you can use a table of values
 - Or for straight lines it can help to rearrange into $y = mx + c$
- Find the point where the lines **intersect**
 - The x and y **solutions** to the simultaneous equations are the x and y **coordinates** of the point of **intersection**
- E.g. To solve $y = x^2 + 3x + 1$ and $y = 2x + 1$ simultaneously
 - First **plot** them both (see graph below)
 - Then find the points of **intersection**, $(-1, -1)$ and $(0, 1)$
 - So the **solutions** are $x = -1$ and $y = -1$ or $x = 0$ and $y = 1$



Your notes



Your notes



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Examiner Tips and Tricks

- When giving your final answer, make sure you indicate which x and y values go together



Worked Example

Solve the equations



Your notes

$$x^2 + y^2 = 36$$
$$x - 2y = 6$$

Number the equations

$$x^2 + y^2 = 36 \quad (1)$$
$$x - 2y = 6 \quad (2)$$

There is one quadratic equation and one linear equation so this must be done by substitution

Equation 2 can be rearranged to make **x** the subject, which can then be substituted into equation 1You could rearrange to make **y** the subject instead, but this results in a fraction which can be more tricky to deal with

Rearranging equation 2

$$x = 2y + 6$$

Substituting into equation 1

$$(2y + 6)^2 + y^2 = 36$$

Expand the brackets

Remember that a bracket squared should be treated the same as double brackets

$$(2y + 6)(2y + 6) + y^2 = 36$$
$$4y^2 + 6(2y) + 6(2y) + 6^2 + y^2 = 36$$

Simplify

$$4y^2 + 12y + 12y + 36 + y^2 = 36$$
$$5y^2 + 24y + 36 = 36$$

Rearrange to form a quadratic equation that is equal to zero

Do this by subtracting 36 from both sides

$$5y^2 + 24y = 0$$

Take out the common factor of **y**

$$y(5y + 24) = 0$$

Solve to find the values of **y** by equating each factor to zero



Your notes

$$y = 0 \text{ or } 5y + 24 = 0$$

Solve the linear equation above

$$y = -\frac{24}{5}$$

So the two y values are

$$y_1 = 0$$

$$y_2 = -\frac{24}{5}$$

Substitute the values of y into one of the equations (the linear equation is easiest) to find the values of x

$$x = 2y + 6$$

$$x_1 = 2(0) + 6 = 6 \qquad x_2 = 2\left(-\frac{24}{5}\right) + 6 = -\frac{18}{5}$$

Write the final solutions in clear pairs

$$\begin{aligned} x_1 &= 6, \quad y_1 = 0 \\ x_2 &= -\frac{18}{5}, \quad y_2 = -\frac{24}{5} \end{aligned}$$